Surd rules

1 Are the following statements true or false?

a $2 - \sqrt{2} = \sqrt{2}$

b $\sqrt{3^2} = (\sqrt{3})^2$

2 Rewrite each of the following expressions in the form $\sqrt{a}$:

a $6\sqrt{7}$ **b** $4\sqrt{14}$ **c** $3\sqrt{5}$ **d** $10\sqrt{10}$

3 Simplify:

a $\dfrac{12 + 9\sqrt{5}}{3}$ **b** $\sqrt{25 \times 2}$ **c** $(\sqrt{45})^2$ **d** $(7\sqrt{15})^2$

e $\sqrt{36x}$ **f** $\dfrac{\sqrt{33}}{\sqrt{11}}$ **g** $\sqrt{\left(1\frac{7}{16}\right)}$

Multiplication and division of surds

4 Simplify the following:

a $\sqrt{2} \times \sqrt{11}$ **b** $2\sqrt{2} \times 2$ **c** $4\sqrt{35} \times \sqrt{5}$ **d** $5\sqrt{7} \times 3\sqrt{5}$

e $\sqrt{5} \times \sqrt{2} \times \sqrt{7}$ **f** $\sqrt{343} \times \sqrt{75}$ **g** $\sqrt{5} \times \sqrt{7}$ **h** $8 \times 10\sqrt{5}$

i $\sqrt{7} \times \sqrt{3} \times \sqrt{11}$ **j** $\sqrt{55} \times \sqrt{11}$ **k** $4\sqrt{11} \times 5$ **l** $2\sqrt{5} \times 15\sqrt{11}$

m $7\sqrt{22} \times \sqrt{2}$ **n** $\sqrt{180} \times \sqrt{48}$ **o** $8\sqrt{15} \times 8\sqrt{5}$ **p** $5\sqrt{17} \times 8\sqrt{3}$

q $17\sqrt{35} \times 4\sqrt{5}$ **r** $8\sqrt{51} \times 9\sqrt{3}$

5 Simplify the following:

a $\sqrt{85} \div \sqrt{17}$ **b** $18\sqrt{7} \div 9$ **c** $24\sqrt{112} \div 3\sqrt{7}$ **d** $8\sqrt{77} \div \sqrt{7}$

e $20\sqrt{14} \div 5\sqrt{2}$ **f** $\sqrt{15} \div \sqrt{5}$ **g** $\sqrt{55} \div \sqrt{5}$ **h** $\sqrt{51} \div \sqrt{17}$

i $\sqrt{21} \div \sqrt{3}$ **j** $\sqrt{91} \div \sqrt{7}$ **k** $40\sqrt{7} \div 8$ **l** $10\sqrt{55} \div \sqrt{11}$

m $15\sqrt{22} \div \sqrt{11}$ **n** $4\sqrt{35} \div 2\sqrt{5}$ **o** $\sqrt{27} \div \sqrt{3}$ **p** $3\sqrt{20} \div \sqrt{5}$

q $5\sqrt{8} \div \sqrt{2}$ **r** $40\sqrt{96} \div 10\sqrt{6}$ **s** $50\sqrt{24} \div 10\sqrt{6}$ **t** $\sqrt{25} \div \sqrt{81}$

u $\sqrt{162} \div \sqrt{8}$

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6 Simplify the following:

a $\sqrt{\frac{28}{7}}$ b $\sqrt{\frac{9}{45}}$ c $\sqrt{\frac{64}{4}}$ d $\sqrt{\frac{48}{144}}$

e $\frac{\sqrt{12}}{\sqrt{36}}$ f $\frac{\sqrt{56}}{\sqrt{14}}$ g $\frac{\sqrt{36}}{\sqrt{81}}$ h $\frac{\sqrt{72}}{\sqrt{32}}$

i $\frac{17\sqrt{21}}{\sqrt{3}}$

Mixed operations with surds

7 Simplify the following:

a $8\sqrt{3}+18\sqrt{3}$ b $-12\sqrt{5}+6\sqrt{5}-12\sqrt{5}$

c $8\sqrt{6}+17\sqrt{13}+19\sqrt{6}-6\sqrt{13}$ d $17\sqrt{6}-8\sqrt{11}+3\sqrt{6}+13\sqrt{11}$

e $\sqrt{6}+\sqrt{54}$ f $\sqrt{180}+\sqrt{500}$

g $\sqrt{32}-\sqrt{98}$ h $4\sqrt{45}+5\sqrt{180}$

8 Use the formula $\sqrt{a}+\sqrt{b}=\sqrt{a+b+2\sqrt{ab}}$ to simplify the following:.

a $\sqrt{3}+\sqrt{12}$ b $\sqrt{8}+\sqrt{2}$ c $\sqrt{18}+\sqrt{8}$ d $2\sqrt{20}+2\sqrt{5}$

9 Expand and simplify:

a $\sqrt{7}\left(-5+\sqrt{5}\right)$ b $7\sqrt{2}\left(\sqrt{7}+6\right)$

c $7\sqrt{11}\left(-6\sqrt{2}+\sqrt{7}\right)$ d $\left(\sqrt{13}+5\right)\left(\sqrt{2}-3\right)$

e $\left(\sqrt{7}-\sqrt{13}\right)\left(\sqrt{2}-\sqrt{11}\right)$ f $\left(\sqrt{11}+1\right)^2$

g $\left(2\sqrt{5}+3\right)^2$ h $\sqrt{11}\left(\sqrt{7}+4\right)$

i $\sqrt{7}\left(3+\sqrt{3}\right)$ j $\sqrt{2}\left(\sqrt{11}-6\right)$

k $3\sqrt{3}\left(\sqrt{13}-5\right)$ l $\sqrt{3}\left(\sqrt{11}+\sqrt{13}\right)$

m $4\sqrt{7}\left(\sqrt{2}-\sqrt{11}\right)$ n $3\sqrt{5}\left(\sqrt{55}+\sqrt{11}\right)$

o $8\sqrt{2}\left(\sqrt{3}-3\sqrt{7}\right)$ p $5\sqrt{2}\left(3\sqrt{5}+4\sqrt{7}\right)$

q $7\sqrt{3}\left(\sqrt{15}+\sqrt{60}\right)$ r $11\sqrt{3}\left(3\sqrt{5}-\sqrt{20}\right)$

s $8\sqrt{11}\left(3\sqrt{7}-4\sqrt{5}\right)$

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10 Simplify the following:

a $3\sqrt{5} \times 8\sqrt{10} \div 2\sqrt{2}$

b $\dfrac{8\sqrt{12} \times 3\sqrt{2}}{4\sqrt{10}}$

c $\dfrac{\sqrt{15} + \sqrt{21}}{\sqrt{3}}$

d $\dfrac{3\sqrt{2} \times 2\sqrt{6}}{\sqrt{8} \times \sqrt{20}}$

11 Find the values of x and y in the following equations:

a $\left(8\sqrt{91} - 5\sqrt{7}\right)^2 = x + y\sqrt{13}$

b $\left(\sqrt{x} + y\right)^2 = 14 - 6\sqrt{5}$

12 Consider the expression $\left(\sqrt{13} + 4\right)^2 + \left(\sqrt{13} + m\right)^2$.

a Expand and simplify the expression.

b Hence, what value of m can be substituted into $\left(\sqrt{13} + 4\right)^2 + \left(\sqrt{13} + m\right)^2$ so that it has a rational value?

Rationalise the denominator

13 State whether the following results in a rational number:

a $\left(\sqrt{8} + \sqrt{3}\right) \times \left(\sqrt{8} + \sqrt{3}\right)$

b $\left(\sqrt{8} + \sqrt{3}\right) \times \left(\sqrt{8} - \sqrt{3}\right)$

c $\left(-\left(\sqrt{8} + 2\right)\right) \times \sqrt{8}$

14 Rationalise the denominator of the following:

a $\dfrac{1}{\sqrt{11}}$

b $\dfrac{\sqrt{2} - 2}{\sqrt{11}}$

c $\dfrac{7}{\sqrt{12}}$

d $\dfrac{14\sqrt{11}}{9\sqrt{13}}$

15 Express the fractions in simplest form with a rational denominator:

a $\dfrac{\sqrt{5}}{\sqrt{15}}$

b $\dfrac{7\sqrt{30}}{\sqrt{10}}$

c $\dfrac{9}{\sqrt{5} - 7}$

d $\dfrac{8}{9\sqrt{11} - 5}$

e $\dfrac{\sqrt{7} + \sqrt{2}}{\sqrt{7} - \sqrt{2}}$

f $\left(\sqrt{10} + 1\right)^{-1}$

g $\dfrac{28}{\sqrt{10} - \sqrt{3}}$

h $\dfrac{\sqrt{7}}{5\sqrt{6} + 4\sqrt{2}}$

16 Express the following in simplest form with rational denominator:

a $\dfrac{1}{\sqrt{2} + 5} + \dfrac{1}{\sqrt{2} - 5}$

b $\dfrac{3}{4\sqrt{5}} - \dfrac{3}{2\sqrt{2}}$

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17 Show that the following expression is a rational number:

$$\frac{3}{\sqrt{2}+3}+\frac{3}{\sqrt{2}-3}$$

18 If $x=2+\sqrt{3}$, simplify and rationalise the denominator of the following expressions:

a $\dfrac{x-\sqrt{3}}{x+\sqrt{3}}$

b $\left(x+\dfrac{1}{x}\right)^2$

19 If $x=4+\sqrt{3}$, simplify and rationalise the denominator of the following expressions:

a x^2

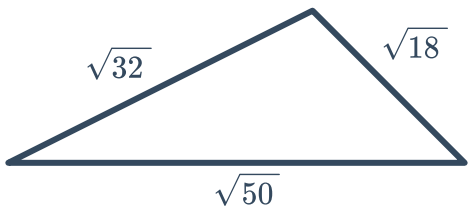
b $\dfrac{x+1}{x-1}$

20 Express the following as a single surd:

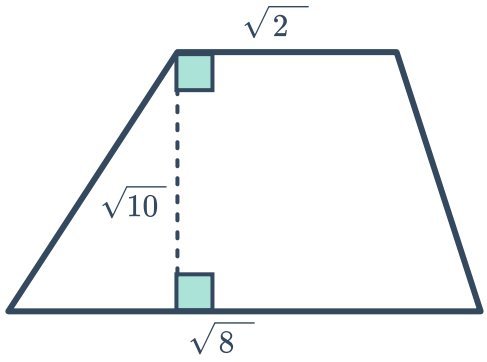
$$42+\frac{6}{7+5\sqrt{2}}$$

Applications

21 Find the perimeter of the triangle in simplified surd form:



22 Find the area of the trapezium in surd form:



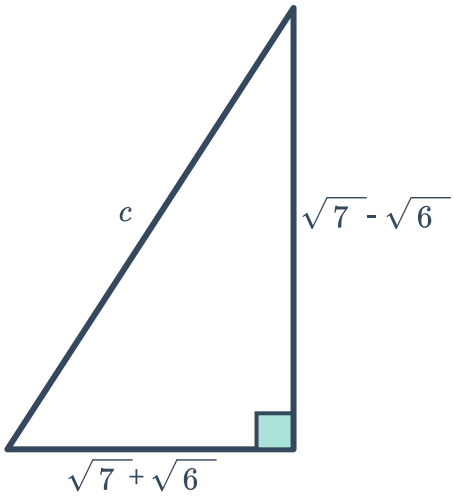
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23 A rectangle has a width of $\left(\sqrt{7}-2\right)$ m and a length of $\left(\sqrt{7}+2\right)$ m.

a Find the exact perimeter of the rectangle.

b Find the exact area of the rectangle.

24 Find c, the length of the hypotenuse of the triangle below. Leave your answer in surd form.



25 Find the exact perpendicular height of a triangle whose area is $36\sqrt{15}$ square centimetres and whose base measures $8\sqrt{5}$ centimetres.

26 A rectangle with height h and width w are believed to be more aesthetically pleasing if their lengths are in the following ratio:

$$\frac{w}{h}=\frac{2}{\sqrt{5}-1}$$

a Rationalise the denominator of the ratio.

b Find a decimal approximation for this ratio correct to the nearest thousandth.

27 A rectangle has a length of $\left(\sqrt{10}+3\right)$ meters and an area of 4 square meters.
What is the exact width of this rectangle? Give your answer with a rational denominator.

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